Curve Sketching – Putting it all together - 1

Follow the algorithm for curve sketching to sketch $f(x) = \frac{2x^2}{x^2-1}$ given that $f'(x) = \frac{-4x}{(x-1)^2(x+1)^2}$ and $f''(x) = \frac{4(3x^2+1)}{(x-1)^3(x+1)^3}$

Domain: $\{x \neq \pm 1\}$ Intercepts: x-int(s): $(0,0)$ y-int $(0,0)$

Asymptote(s):

Horizontal

$y = 2$

Crosstest

$2 = \frac{2x^2}{x^2-1}$

Does not cross H.A.

$f(100) = \frac{20000}{9999} > 2$

$\Rightarrow$ above H.A.

$f(-100) = \frac{20000}{9999} > 2$

$\Rightarrow$ above H.A.

Vertical $x = 1$ $x = -1$

$\lim_{x \to 1^-} \frac{2x^2}{x^2-1} = -\infty$ $\lim_{x \to -1^-} \frac{2x^2}{x^2-1} = \infty$

$\lim_{x \to 1^+} \frac{2x^2}{x^2-1} = \infty$ $\lim_{x \to -1^+} \frac{2x^2}{x^2-1} = -\infty$

First Derivative Test

Critical number(s): $0, \pm 1$ Local Extrema: Maxima: $(0,0)$ Minima: none

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
-4	$-$	$-$	$-$	$-$
x	$-$	$-$	$+$	$+$
$(x-1)^2$	$+$	$+$	$+$	$+$
$(x+1)^2$	$+$	$+$	$+$	$+$
$f'(x)$	$+$	$+$	$-$	$-$
$f(x)$	$\uparrow$	$\uparrow$	$\downarrow$	$\downarrow$

V.A $x = -1$ Max $(0,0)$ V.A. $x = 1$

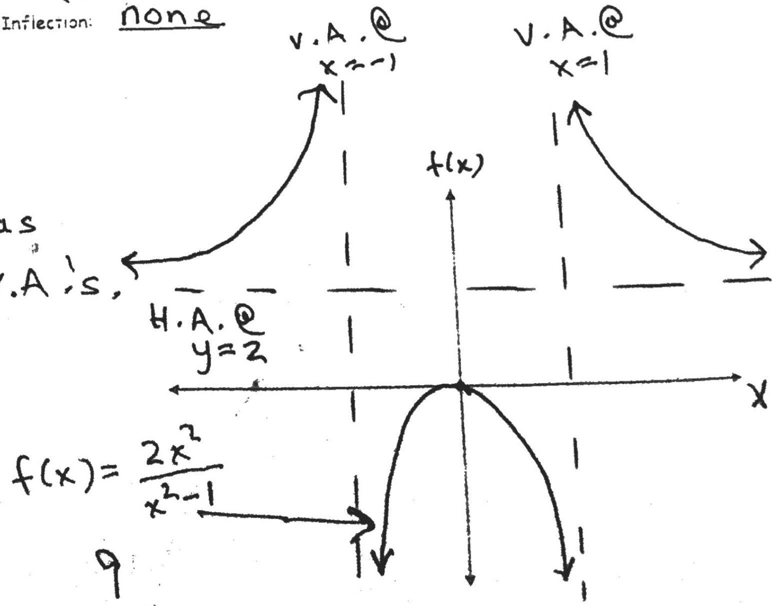
Test for Concavity Point(s) of Inflection: none

possible poi

@ $x = \pm 1$

Not possible as

$x = \pm 1$ are V.A's.



Sketch of Curve

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Curve Sketching – Putting it all together - 2

Follow the algorithm for curve sketching to sketch $f(x) = \frac{x^2-16}{x^2-4}$ given that $f'(x) = \frac{24x}{(x-2)^2(x+2)^2}$ and $f''(x) = \frac{-24(3x^2+4)}{(x-2)^3(x+2)^3}$

Domain: $\{x \neq \pm 2\}$ Intercepts: x-int(s): $(-4,0), (4,0)$ y-int $(0,4)$

Asymptote(s):

Horizontal $y = 1$

$f(100) < 1$

$\Rightarrow$ below H.A.

$f(-100) < 1$

$\Rightarrow$ below H.A.

Vertical $x = 2$

$\lim_{x \to 2^-} \frac{x^2-16}{x^2-4} = \infty$

$\lim_{x \to 2^+} \frac{x^2-16}{x^2-4} = -\infty$

$x = -2$

$\lim_{x \to -2^-} \frac{x^2-16}{x^2-4} = -\infty$

$\lim_{x \to -2^+} \frac{x^2-16}{x^2-4} = \infty$

First Derivative Test

Critical number(s): $0, \pm 2$ Local Extrema: Maxima: none Minima: $(0,4)$

Interval	$(-\infty,-2)$	$(-2,0)$	$(0,2)$	$(2,\infty)$
x	$-$	$-$	$+$	$+$
$(x-2)^2$	$+$	$+$	$+$	$+$
$(x+2)^2$	$+$	$+$	$+$	$+$
$f'(x)$	$-$	$-$	$+$	$+$
$f(x)$	$\downarrow$	$\downarrow$	$\uparrow$	$\uparrow$

min @ $(0,4)$

Test for Concavity Point(s) of Inflection: none

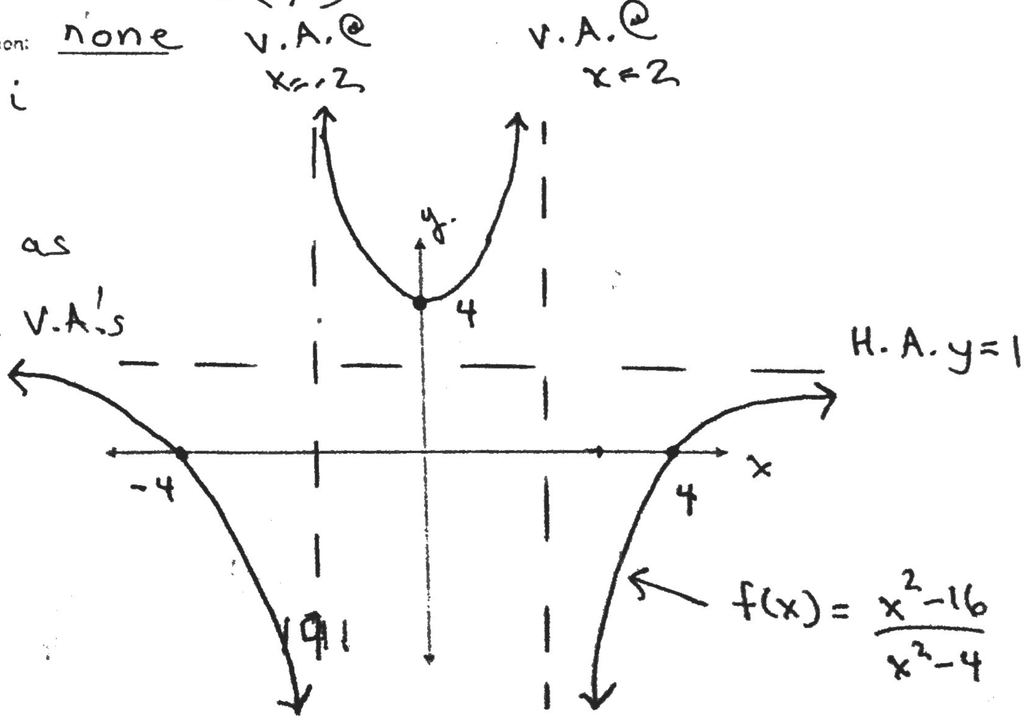
possible poi

@ $x = \pm 2$

not possible as

$x = \pm 2$ are V.A.'s

Sketch of Curve



Curve Sketching – Putting it all together - 3

Follow the algorithm for curve sketching to sketch $f(x) = x^{\frac{2}{3}}(x-5)$ given that $f'(x) = \frac{5(x-2)}{3\sqrt[3]{x}}$ and $f''(x) = \frac{10(x+1)}{9x^{\frac{4}{3}}}$

Domain: $(-\infty, \infty)$ Intercepts: x-int(s): $(0,0), (5,0)$ y-int $(0,0)$

Asymptote(s):

Horizontal: NONE

Vertical: NONE

First Derivative Test

Critical number(s): 2, 0 Local Extrema: Maxima $(0,0)$ Minima: $(2, -4.76)$

Interval	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$x-2$	−	−	+
$\sqrt[3]{x}$	−	+	+
$f'(x)$	+	−	+
$f(x)$	↑	↓	↑

cusp

Min @ $(2, -4.76)$

Max @ $(0, 0)$

Test for Concavity Point(s) of Inflection: $(-1, -6)$

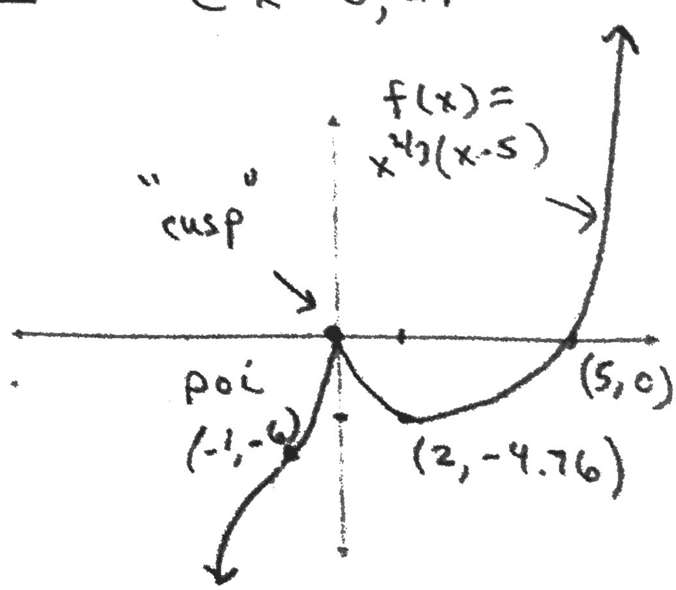
possible poi @ $x = 0, -1$

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, \infty)$
$x+1$	−	+	+
$x^{4/3}$	+	+	+
$f''(x)$	−	+	+
$f(x)$	c.D.	c.u.	c.u.

Sketch of Curve

poi at $(-1, -6)$ cusp

11.



Curve Sketching – Putting it all together - 4

Follow the algorithm for curve sketching to sketch $f(x) = (x-1)^3(x+1)^3$ given that
$) = 6x(x-1)^2(x+1)^2$ and $f''(x) = 6(x-1)(x+1)(5x^2-1)$

Domain: $(-\infty, \infty)$

Intercepts: x-int(s): $(1,0), (-1,0)$ y-int $(0,-1)$

Asymptote(s):
Horizontal none

Vertical none.

First Derivative Test

Critical number(s): $0, \pm 1$

Local Extrema: Maxima: none Minima: $(0,-1)$

Interval	$(-\infty,-1)$	$(-1,0)$	$(0,1)$	$(0,\infty)$
x	−	−	+	+
$(x-1)^2$	+	+	+	+
$(x+1)^2$	+	+	+	+
$f'(x)$	−	−	+	+
$f(x)$	↓	↓	↑	↑

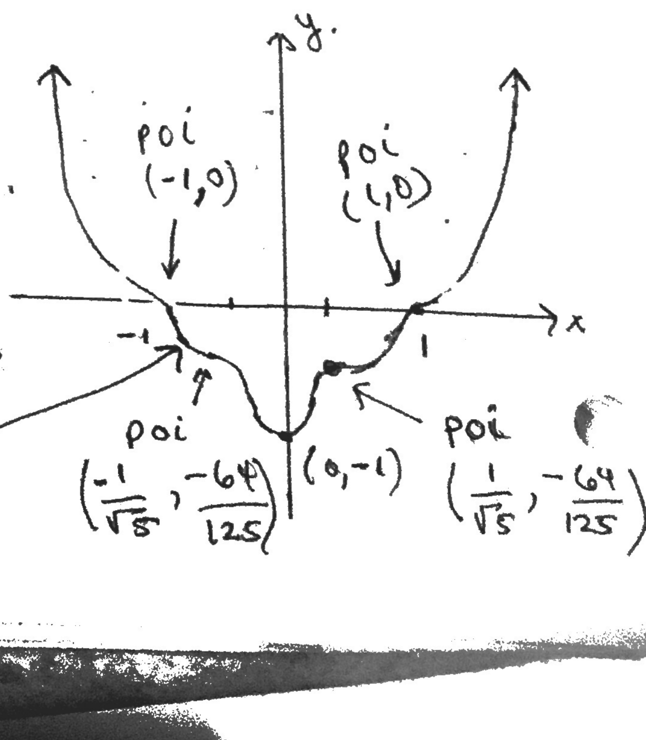
Min @
$x = 0 \rightarrow (0,-1)$

Test for Concavity Point(s) of Inflection: $(-1,0), \left(\frac{-1}{\sqrt{5}}, \frac{-64}{125}\right), \left(\frac{1}{\sqrt{5}}, \frac{-64}{125}\right), (1,0)$

Interval	$(-\infty,-1)$	$\left(-1,-\frac{1}{\sqrt{5}}\right)$	$\left(-\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$	$\left(\frac{1}{\sqrt{5}}, 1\right)$	$(1,\infty)$
$x-1$	−	−	−	−	+
$x+1$	−	+	+	+	+
$x-\frac{1}{\sqrt{5}}$	−	−	−	+	+
$x+\frac{1}{\sqrt{5}}$	−	−	+	+	+
$f''(x)$	+	−	+	−	+
$f(x)$	C.U	C.D.	C.U.	C.D.	C.U.

Sketch of Curve

$f(x) = (x-1)^3(x+1)^3$



Curve Sketching – Putting it all together - 5

Follow the algorithm for curve sketching to sketch $f(x) = x\sqrt{8-x^2}$ given that $f'(x) = \frac{-2(x-2)(x+2)}{\sqrt{8-x^2}}$ and $f''(x) = \frac{2x(x+2\sqrt{3})(x-2\sqrt{3})}{(x^2+8)^{\frac{3}{2}}}$

Domain: $[-2\sqrt{2}, 2\sqrt{2}]$ Intercepts: x-int(s): $(0,0), (\pm 2\sqrt{2})$ y-int $(0,0)$

Asymptote(s):

Horizontal

NONE

Vertical

NONE

First Derivative Test

Critical number(s): $\pm 2, \pm 2\sqrt{2}$ Local Extrema: Maxima: $(2, 4)$ Minima: $(-2,-4)$

Interval	$(-\infty,-2\sqrt{2})$	$(-2\sqrt{2},-2)$	$(-2,2)$	$(2,2\sqrt{2})$	$(2\sqrt{2},\infty)$
-2	−	−	−	−	−
$x-2$	−	−	−	+	+
$x+2$	−	−	+	+	+
$\sqrt{8-x^2}$	DNE	+	+	+	DNE
$f'(x)$		−	+	−	
$f(x)$		↓	↑	↓	

$\therefore$ min @ $(-2,-4)$

Max @ $(2, 4)$

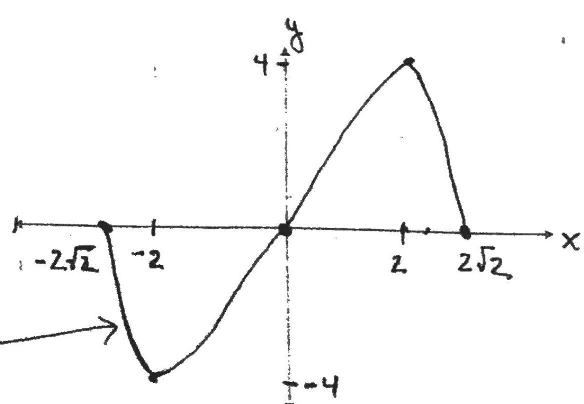
Test for Concavity Point(s) of Inflection: $(0,0)$

Interval	$(-\infty,-2\sqrt{3})$	$(-2\sqrt{3},0)$	$(0,2\sqrt{3})$	$(2\sqrt{3},0)$
x		−	+	
$x+2\sqrt{3}$		+	+	
$x-2\sqrt{3}$	DNE	−	−	DNE
$(x^2+8)^{\frac{3}{2}}$		+	+	
$f''(x)$		+	−	
$f(x)$		C.U.	C.D.	

Sketch of Curve

$\therefore$ poi @ $(0,0)$

$f(x)=x\sqrt{8-x^2}$



Curve Sketching – Putting it all together - 6

Follow the algorithm for curve sketching to sketch $f(x) = (x-3)^2 \cdot \sqrt[3]{x+5} + 5$ given that $f'(x) = \frac{(x-3)(7x+27)}{3(x+5)^{\frac{2}{3}}}$ and $f''(x) = \frac{4(7x^2+54x+63)}{9(x+5)^{\frac{5}{3}}}$

Domain: $(-\infty, \infty)$ Intercepts: x-int(s): difficult to calculate y-int $(0, 20.4)$

Asymptote(s):

Horizontal: none

Vertical: none.

First Derivative Test

Critical number(s): $3, -\frac{27}{7}, -5$ Local Extrema: Maxima: $\left(-\frac{27}{7}, 54.16\right)$ Minima: $(3, 5)$

Interval	$(-\infty, -5)$	$(-5, -\frac{27}{7})$	$(-\frac{27}{7}, 3)$	$(3, \infty)$
$x-3$	−	−	−	+
$7x+27$	−	−	+	+
$(x+5)^{2/3}$	+	+	+	+
$f'(x)$	+	+	−	+
$f(x)$	↑	↑	↓	↑

Max @ $\left(-\frac{27}{7}, 54.16\right)$

Min @ $(3, 5)$

Test for Concavity Point(s) of Inflection: $(-6.28, -88.6), (-5, 5), (-1.43, 35.03)$

Interval	$(-\infty, -6.28)$	$(-6.28, 5)$	$(-5, -1.43)$	$(-1.43, \infty)$
$x+1.43$	−	−	−	+
$x+6.28$	−	+	+	+
$(x+5)^{5/3}$	−	−	+	+
$f''(x)$	−	+	−	+
$f(x)$	C.D.	C.U.	C.D.	C.U.

Sketch of Curve

$f(x) = (x-3)^2 \sqrt[3]{x+5} + 5$